

TUGAS 1
MATA KULIAH
Dasar Pemrograman R / STIK4111



DISUSUN OLEH

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1. Penjelasan perintah R

a. `ls(pat = "y")`

Perintah ini digunakan untuk menampilkan nama objek pada environment yang mengandung huruf "y".

Pada simulasi awal, hasilnya `character(0)` karena belum ada objek yang cocok.

b. `for (i in 1:25) { print(i) }`

Perintah ini menampilkan bilangan 1 sampai 25 secara berurutan.

c. `setwd("C:/B")`

Perintah ini digunakan untuk mengubah working directory ke folder C:/B.

Pada awalnya perintah ini dapat menimbulkan error jika folder C:/B belum tersedia. Setelah folder tersebut dibuat, perintah `setwd("C:/B")` berhasil dijalankan dan `getwd()` menampilkan working directory "C:/B".

The image shows a screenshot of the RGui - [Untitled - R Editor] window. The window has a menu bar with 'File', 'Edit', 'Packages', 'Windows', and 'Help'. Below the menu bar is a toolbar with icons for file operations and running code. The main text area contains the following R code:

```
x <- 10
y <- 20
umur <- 22
miles <- 90

ls(pat = "y")

for (i in 1:25) {
  print(i)
}

setwd("C:/B")
getwd()
```

2. Perintah R untuk membuat data frame mahasiswa

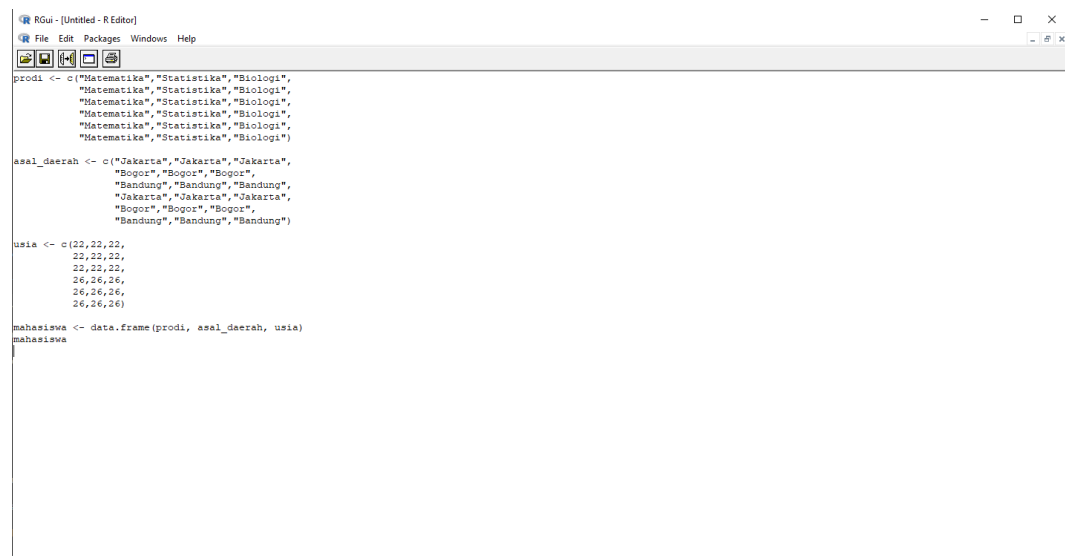
```
prodi <- rep(c("Matematika", "Statistika", "Biologi"), 6)
```

```
asal_daerah <- rep(c("Jakarta", "Bogor", "Bandung"), each = 3, times = 2)
```

```
usia <- c(rep(22, 9), rep(26, 9))
```

```
mahasiswa <- data.frame(prodi, asal_daerah, usia)
```

mahasiswa



```
RGui - [Untitled - R Editor]
File Edit Packages Windows Help

prodi <- c("Matematika", "Statistika", "Biologi",
           "Matematika", "Statistika", "Biologi",
           "Matematika", "Statistika", "Biologi",
           "Matematika", "Statistika", "Biologi",
           "Matematika", "Statistika", "Biologi")

asal_daerah <- c("Jakarta", "Jakarta", "Jakarta",
                 "Bogor", "Bogor", "Bogor",
                 "Bandung", "Bandung", "Bandung",
                 "Jakarta", "Jakarta", "Jakarta",
                 "Bogor", "Bogor", "Bogor",
                 "Bandung", "Bandung", "Bandung")

usia <- c(22, 22, 22,
          22, 22, 22,
          22, 22, 22,
          26, 26, 26,
          26, 26, 26,
          26, 26, 26)

mahasiswa <- data.frame(prodi, asal_daerah, usia)
mahasiswa
```

3. Data aktivitas harian, matriks 4x5, dan ukuran pemusatan

a. Contoh data waktu aktivitas harian (dalam menit): 180, 420, 30, 90, 150, 60, 80, 40, 30, 120, 45, 50, 100,

70, 60, 120, 90, 40, 35, 60.

b. Data tersebut dibuat menjadi matriks 4x5 menggunakan fungsi `matrix()`.

```
waktu <- c(180, 420, 30, 90, 150, 60, 80, 40, 30, 120, 45, 50, 100, 70, 60,
120, 90, 40, 35, 60)
mat_waktu <- matrix(waktu, nrow = 4, ncol = 5, byrow = TRUE)
mat_waktu
min(mat_waktu)
max(mat_waktu)
mean(mat_waktu)
median(mat_waktu)
modus <- as.numeric(names(sort(table(mat_waktu), decreasing = TRUE)[1]))
modus
```

c. Hasil perhitungan di R:

- Nilai minimum = 30
- Nilai maksimum = 420
- Rata-rata = 93.5
- Median = 65
- Modus = 60

The image shows a screenshot of the RGui application window. The title bar reads "RGui - [Untitled - R Editor]". The menu bar includes "File", "Edit", "Packages", "Windows", and "Help". Below the menu bar is a toolbar with icons for file operations and editing. The main text area contains the following R code:

```
waktu <- c(120, 420, 30, 60, 180,
          90, 45, 60, 300, 120,
          480, 40, 70, 100, 50,
          200, 35, 55, 80, 65)

mat_waktu <- matrix(waktu, nrow = 4, ncol = 5, byrow = TRUE)
mat_waktu

min(mat_waktu)
max(mat_waktu)
mean(mat_waktu)
median(mat_waktu)

modus <- as.numeric(names(sort(table(mat_waktu), decreasing = TRUE)[1]))
modus
```

The output of the code is not visible in the screenshot.